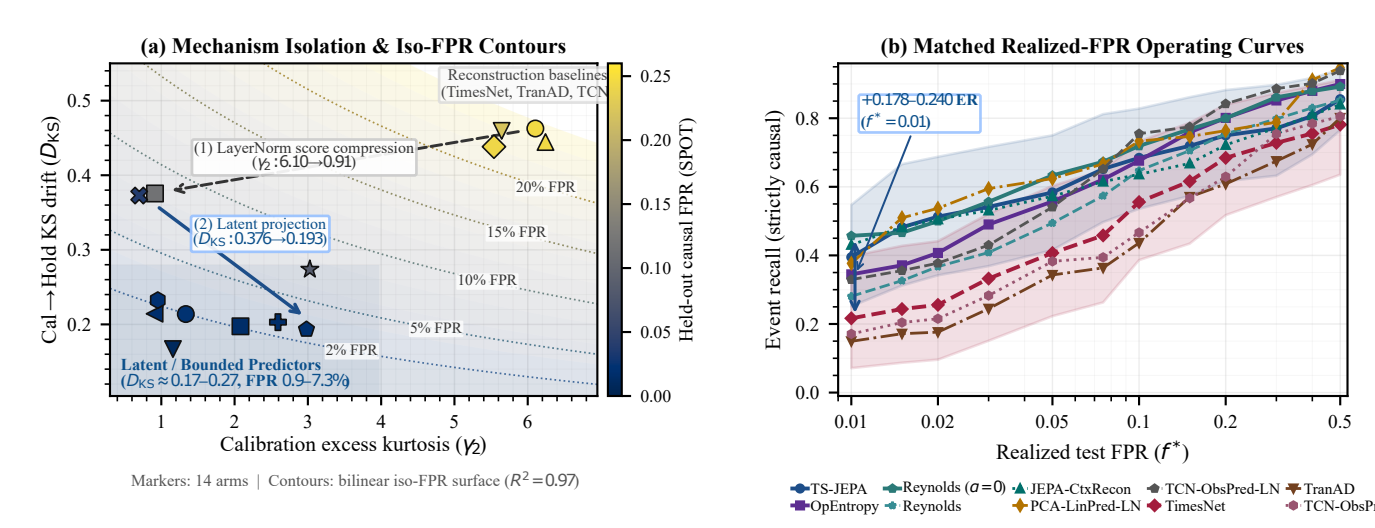

(a) Mechanism Isolation & Iso-FPR Contours
Reconstruction baselines
(TimesNet, TranAD, TCN)
(1) LayerNorm score compression
(γ_2: 6.10→0.91)
(2) Latent projection
(D_{KS}: 0.376→0.193)
20% FPR
15% FPR
10% FPR
5% FPR
2% FPR
Latent / Bounded Predictors
(D_{KS} ≈ 0.17–0.27, FPR 0.9–7.3%)
Cal→Hold KS drift (D_{KS})
Calibration excess kurtosis (γ_2)
Held-out causal FPR (SPOT)
Markers: 14 arms | Contours: bilinear iso-FPR surface ($R^2 = 0.97$)
(b) Matched Realized-FPR Operating Curves
+0.178–0.240 ER
($f^* = 0.01$)
Event recall (strictly causal)
Realized test FPR (f^*)
TS-JEPA
Reynolds ($\sigma = 0$)
JEPA-CtxRecon
TCN-ObsPred-LN
TranAD
OpEntropy
Reynolds
PCA-LinPred-LN
TimesNet
TCN-ObsPr